

# Luis Fernando Palacios Flores

✉ [lf.palaciosf@gmail.com](mailto:lf.palaciosf@gmail.com)  
☎ +39 350 934 3312

🌐 [luisfpal.github.io](https://luisfpal.github.io)  
✉ [luisfpal](#)

🔗 [luisfpal](#)  
📄 [luisfpal](#)

## Education

**M.Sc. in Data Science and Artificial Intelligence**  
*University of Trieste, Trieste, Italy*

Sep 2023 – Sep 2025  
GPA: 110/110 cum laude

**B.Sc. in Physics**  
*Escuela Politécnica Nacional, Quito, Ecuador*

Apr 2017 – Nov 2022  
GPA: 26.10/30

**Master in Data Management and Curation (postgraduate course)**  
*SISSA and Area Science Park, Trieste, Italy*

Sep 2025 – Jun 2026

## Research

**Master's Thesis**, Laboratory of Data Engineering, Area Science Park

Mar 2025 – May 2026

- Showed that visual instruction tuning places image features in the middle layers of the language model. Tuning layers 6–16 of LLaVA-7B reached 98.8% of full fine-tuning at 86.1% of the training time. Tuning layers 8–25 of OneVision-4B reached 100.4% at 75.7% of the time.
- **NeurIPS 2026, poster.** Luis Palacios\*, Lorenzo Basile\*, Diego Doimo, Alberto Cazzaniga. [Visual Instruction Tuning Aligns Modalities through Abstraction](#). 📄 Equal contribution with Lorenzo Basile. Basile, Doimo, and Cazzaniga designed the experiments. I wrote the code and ran them with Basile and Doimo.

**Data Management and Curation**, Area Science Park

Nov 2025 – Jun 2026

- Built two independent services and deployed both. Buckets Explorer lets people govern where files live. 🔄 🔗 📄
- Built an API that takes an image and returns a classification. The working example is scanning-electron-microscope images. The two services were not joined. 🔄 📄
- **IEEE eScience 2026, not accepted.** With Federica Bazzocchi and Tommaso Rodani. The review wanted throughput experiments the paper did not have. 📄 🔗

**Bachelor's Thesis**, Escuela Politécnica Nacional

2022

- Compared magnetic spectra of the Earth and the Sun after separating the signals with empirical mode decomposition. 📄 🔄

## Coursework

---

- Networks that treat a rotated image as the same object. Group-equivariant CNNs. 
- Two models that turn a sentence into an image. 
- Whether a vision model's explanation survives an attack on the image. 
- A simulated walker and a humanoid, taught to move. SAC and TD3. 
- A diffusion model that repairs categorical data. 
- Strategies for a repeated game, the iterated prisoner's dilemma. 
- A file store people can sign in to, watched under traffic. Nextcloud. 
- How a cluster sends one message to every machine, and how it waits for all of them. 
- With Piero Zappi and Marco Tallone, a similar share. Sparse matrices, a C++ toolbox, and Python bindings. 
- Four people. Who buys health insurance, when almost every answer is no. 
- Two papers, with small code: a physical system from rough and precise data, and PPO. 

## Work Experience

---

Teaching Technician, EPN, Quito, Ecuador

May - Aug 2023

- Supervised physics laboratory sessions and assessed the lab reports.

## Skills

---

**Programming:** Python, PyTorch, C++, Kubernetes, Git, Linux

**Languages:** Spanish (native), English (C1), Italian (A1)

## Extracurricular Activities and Achievements

---

**Research & Study Fellowships** — MDMC — LADE research fellowship; SISSA fee scholarship 2025 - 2026

**Merit-Based Study Fellowship** — DSAI — SISSA; Collegio Fonda (Trieste, Italy) 2023 - 2025

**Vice-President of the Theoretical Physics Club** , EPN Nov 2021 - Mar 2023

- Organized physics and programming workshops and maintained club GitHub repositories. 

## References

---

Dr. Alberto Cazzaniga, thesis advisor. Dr. Diego Doimo, thesis advisor.

References available on request.